

# RUPAK RAUT

[rupakraut78@gmail.com](mailto:rupakraut78@gmail.com) — [linkedin.com/in/rupak-raut](https://www.linkedin.com/in/rupak-raut) — [github.com/rupaut98](https://github.com/rupaut98)

## EDUCATION

**The University of Southern Mississippi**

*Graduated:* May 2026

*Major:* Computer Science & Mathematics

*GPA:* 4.0

*Coursework:* Data Structures & Algorithms, Machine Learning, Databases, Operating Systems, Linear Algebra

## WORK EXPERIENCE

**Software Engineer**

Nov 2025 - Present

*Retina Specialists of Mississippi*

- Owned end-to-end delivery of a full-stack TypeScript platform (Next.js, React, PostgreSQL) for an enterprise client, scoping requirements on-site and extending it to their workflows, now live across 7 locations.
- Integrated the platform with the client's existing enterprise systems (Athena and Nextech) over OAuth 2.0, syncing data into their day-to-day operations, and served as the technical point of contact through launch and beyond.
- Engineered a type-safe PostgreSQL data layer with Drizzle ORM and Zod, enforcing 100+ row-level-security policies, AES-256-GCM encryption, audit logging, and atomic transactions.
- Built dynamic intake, e-signature, and PDF flows plus two companion apps, a staff-training platform and a locked-down Android kiosk, safeguarded by a Vitest, Playwright, and CI pipeline.

**Software Engineer**

May 2025 - Present

*IllumibotAI*

- Built 40+ Node.js/TypeScript serverless functions orchestrating 10+ AI providers (Gemini, Runway, Veo) through event-triggered pipelines with atomic token transactions.
- Shipped the consumer app in Flutter across iOS, Android, and web, including a self-built multi-timeline video editor with FFmpeg export that drove thousands in revenue.
- Built a FastAPI and OpenCV microservice for real-time image processing, with native iOS and Android plugins for adaptive performance on long exports.

**Founding Engineer**

Feb 2024 - Dec 2024

*OmniSynkAI*

*Top 10% YC Applicant (W24), Techstars Finalist, Angel-Funded*

- Led a team of 5 to build a B2B e-commerce marketplace from 0 to 1, automating product listings, order fulfillment, and inventory sync across multiple online marketplaces.
- Owned the backend and data layer in Node.js (Fastify), PostgreSQL, and Prisma, deployed on GCP with GitHub Actions CI/CD.
- Fine-tuned a BERT model on 100K+ product entries to 95% accuracy for catalog auto-categorization, and cut backend response time 30% with Redis caching.

## PROJECTS

**FreeFEM-CodeGen**

- Built a two-stage QLoRA pipeline (4-bit NF4, rsLoRA) that fine-tunes a 7B code model into a FreeFEM finite-element solver specialist, lifting execution pass rate from 0% to 80% and beating zero-shot Claude Sonnet 4.5, Gemini 3 Flash, and a 123B model 17x its size.
- Built an execution-grounded eval harness that compiles and runs every generated program through the real solver, reaching 98% pass@10 with a self-repair loop and multi-seed ablations across three model architectures on H100 and H200 GPUs.

**Neuro-Symbolic LLM Validator**

*Radiance Technologies Innovation Bowl Finalist*

- Built a neuro-symbolic hallucination-detection pipeline that extracts atomic claims with FActScore-style voting, plans a provably complete verification order with a Fast Downward PDDL planner, and runs 15 evidence-gathering agents over a ChromaDB RAG corpus.
- Engineered a cross-provider validation layer calling Gemini, GPT-5, and Claude over raw aiohttp, scoring each claim through a deterministic 9-signal scorer that graded 97 claims at 84% cross-validator agreement in under 5 minutes per run.

**Sprig (Rust)**

- Built a sparse n-gram indexed code-search engine in Rust that delivers sub-millisecond regex search across large codebases.

## OPEN-SOURCE CONTRIBUTIONS

- [ByteDance trae-agent](#) (11k+ stars) — Merged 12 PRs adding Gemini and OpenAI (GPT-4o/o3) provider integrations and a JSONPath-based edit tool with 17 unit tests.
- [Unsloth](#) (28k+ stars) — Patched a mixed-precision training crash (FP16-to-FP32 embedding upcast) and a Gemma 3 vision processor bug.

## SKILLS

**Languages:** TypeScript, JavaScript, Python, Rust, SQL

**Full-Stack:** React, Next.js, Node.js, Fastify, REST APIs, PostgreSQL, Prisma, Drizzle ORM

**Integrations & AI:** OAuth 2.0, third-party and ERP integrations, LLM fine-tuning, agentic pipelines, PyTorch

**DevOps & Testing:** Docker, GCP, GitHub Actions, Vitest, Playwright